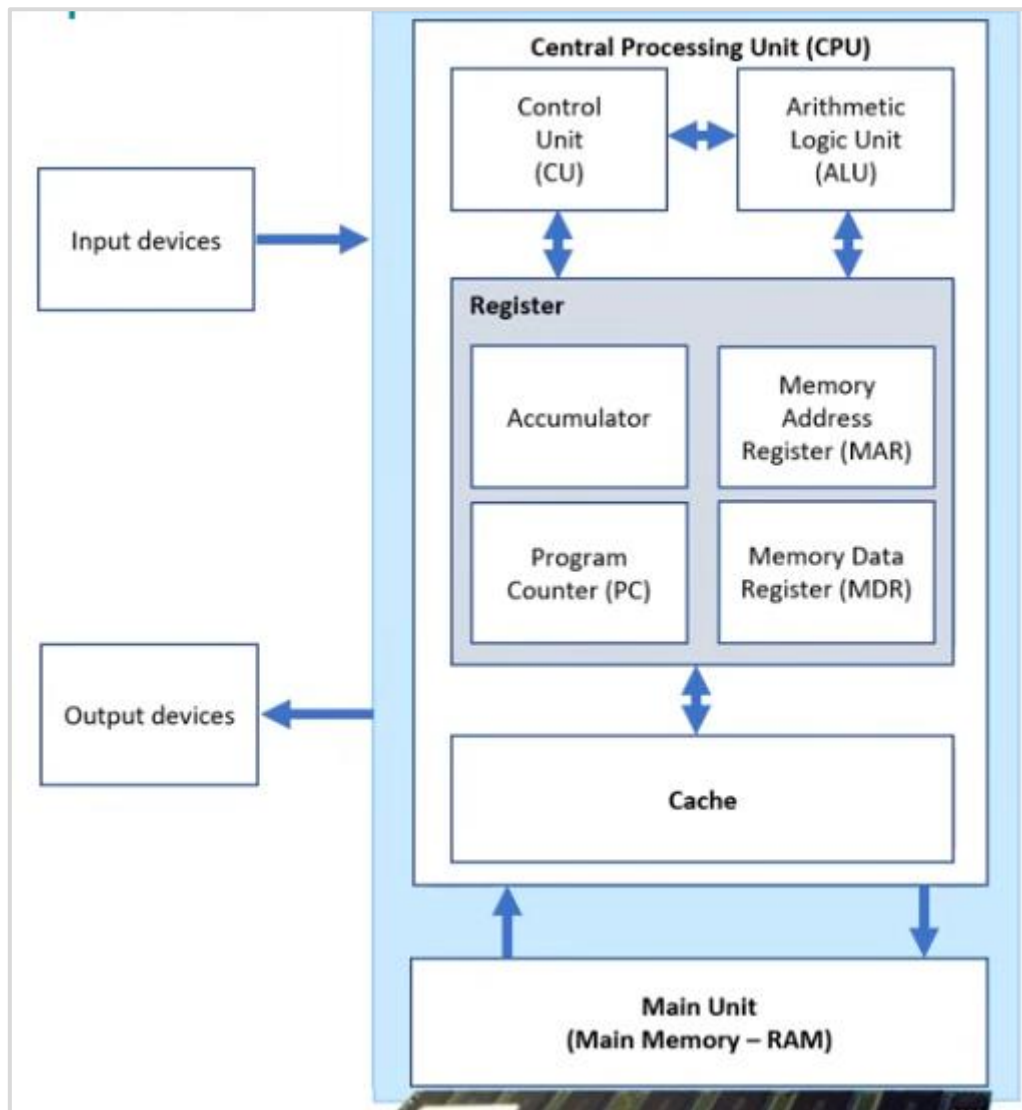


Question 1: Describe what happens during the fetch-execute cycle.

- Data and instructions are **fetch**ed from memory, RAM, or primary storage.
- Data and instructions are **stored using the registers** (e.g., a specific register storing an address or data).
- Data and instructions are **dec**oded (often split into opcode and operand).
- Data and instructions are **exec**uted or processed.
- The **ALU** performs logical and arithmetic calculations during the execution phase.

Answer:

- Data and instructions are **fetch**ed from memory (RAM),
- **dec**oded (split into opcode and operand),
- and **exec**uted by the CPU, with the ALU performing calculations



Question 2 (i): Identify component(s) matching the statements.

- **Stores a single address:** Memory Address Register (MAR) and Program Counter.
- **Stores frequently used instructions:** Cache.
- **Is a register:** MAR, Memory Data Register (MDR), and Program Counter.
- **Stores all currently running data and instructions:** RAM.

i. The table has **five** components of a computer, and **four** statements.

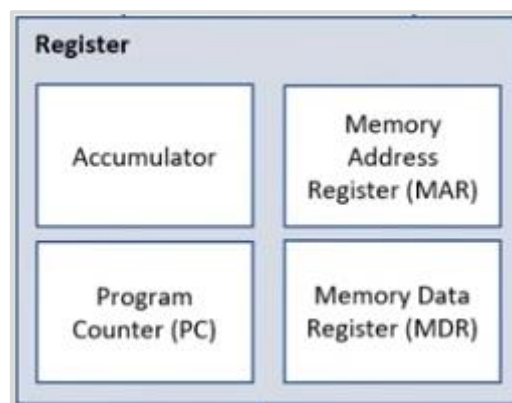
Tick (✓) **one or more** boxes in each row to identify which component(s) each statement describes.

Statement	MAR	MDR	Cache	Program Counter	RAM
It stores a single address					
It stores frequently used instructions					
It is a register					
It stores all currently running data and instructions					

[4]

Question 2 (ii): Identify one register not given in part (i) and its purpose.

- **Register:** Accumulator.
- **Purpose:** To store the **results** of arithmetic and logical operations.



Question 3: Provide the name or definition for CPU components.

- **Program Counter (PC):** Stores the address of the **next** instruction to be fetched from memory and increments each cycle.
- **Control Unit (CU):** Sends signals to **synchronise**, control, or coordinate the processor, hardware, and the flow of data.
- **Memory Address Register (MAR):** Stores the address of the data to be **fetched from**, or the address where data is to be **stored**.
- **Arithmetic Logic Unit (ALU):** Performs the mathematical and logical calculations.

Question 4: Reason why cache size affects CPU performance.

- Cache is **faster** to transfer data to and from than RAM.
- This increased transfer speed makes the CPU **more efficient**.

Question 5: Why is the cruise control an embedded system?

- It has a **specific purpose** or performs a limited, dedicated task.
- It is **built within a larger device** (the car).
- It uses its own **dedicated hardware** or sensors and a microprocessor with built-in firmware/ROM.

Question 6: Why a 3.2 GHz processor is faster than a 1.2 GHz processor.

- It has a **higher clock speed**.
- It can run **more Fetch-Execute cycles per second**, allowing more instructions to be executed in that time.

Question 7 (i): Why a computer **needs both primary and secondary storage.**

- **Primary storage** (RAM) is needed to store **active** data, software, and the OS that the processor needs to access; without it, the computer cannot start or work.
- **Secondary storage** is needed to store files **permanently or long-term** so they are not lost when the power is turned off.

Question 7 (ii): Example of secondary storage and data stored.

- **Device:** Hard drive, SSD, USB stick, or CD/DVD.
- **Data:** Software, images, documents, or data moved from RAM to virtual memory.

Question 7 (iii): Correcting Virtual Memory (VM) statements.

1. A section of **secondary storage** is partitioned to act as VM.
2. Data from **RAM** is transferred into VM.
3. (True) VM is needed when RAM is full.
4. Data from VM is transferred back to **RAM** when needed.

Question 8 (a)(i): Define the term pixel.

- The **smallest unit** or part of an image.
- A single square, dot, or diode that has **one single color**.

Question 8 (a)(ii): Total file size of **10 images (800 x 500, **8-bit**) in **kilobytes**.**

- Working: $(800 * 500 * 10 * 8) / 8 / 1000 = 4000$ **kilobytes**. (Note: dividing by 1024 is also acceptable, resulting in 3906.25 KB).

Question 8 (a)(iii): Minimum bits needed for 240 colors.

- **8 bits** (because 2^8 provides 256 possible values, which covers 240).

Question 8 (b): Choosing storage type for moving files.

- **Choice:** Solid-state.
- **Justification:** It is **durable** (no moving parts), **portable** (small and lightweight), and has a **large capacity** to store many large image files.

Question 9 (a): Base Conversions.

- 38 (Denary) = **00100110** (Binary) = **26** (Hex).
- **78** (Denary) = 01001110 (Binary) = **4E** (Hex).
- 156 (Denary) = **10011100** (Binary) = **9C** (Hex).
- **215** (Denary) = **11010111** (Binary) = **D7** (Hex).

Question 9 (b): Binary addition of 01110011 and 00110100.

- Answer: **10100111** (marks are awarded for showing the 3 carries in the working).

Question 9 (c): 2-place right binary shift on 01110111.

- Answer: **00011101**.

Question 10: Character Sets description.

- A character set stores **all** characters. Each is given a **unique** binary code. Upper and lower-case letters are given **unique (or different)** codes. ASCII uses **8 bits** per character. If 'F' is 70, then 'L' is **76**.

Question 11 (a): Convert Hex 2F to denary.

- Working: $(2 * 16) + 15 =$ **47**.

Question 11 (b): Convert Binary 10110000 to Hex.

- Answer: **B0**.

Question 12 (i): Identify correct sound sampling description.

- The **amplitude** of the wave is measured at **set intervals**.

Question 12 (ii): Effect of changing bit depth.

- Changing the bit depth changes the **number of bits per sample**.
- It affects the **file size** and the **accuracy/quality** of the recording (increasing bit depth allows a wider range of amplitudes to be measured).

Question 13 (i): Two benefits of compressing data for email.

- Reduces **file size**.
- Ensures the email **transmits faster** (uploading and downloading) and uses less bandwidth.

Question 13 (ii): Why lossy compression is inappropriate for text files.

- Lossy compression results in **permanent data loss**.
- Text files would become **corrupted** and unreadable because the data cannot be recovered.